

Inputs / Outputs / Integration Notes

(a) What data does the model receive? (Inputs)

Manual Entry

- Patient age and sex
- Presenting complaint
- Pain score
- Nurse observations
- Additional symptoms or clinical notes

Electronic Health Record (EHR) Pull

- Patient demographics
- Medical history
- Allergies
- Previous ED visits
- Current medications
- Existing diagnoses (where available)

Device Stream

- Heart rate
- Blood pressure
- Respiratory rate
- Oxygen saturation (SpO₂)
- Body temperature

(b) What does the model emit? (Outputs)

The AI-assisted triage system generates:

- Predicted **Emergency Severity Index (ESI) Level (1–5)**
- Confidence score for the prediction
- Colour-coded urgency alert
 -  Red – Immediate
 -  Orange – Emergent
 -  Yellow – Urgent
 -  Green – Less Urgent
- Key clinical factors contributing to the recommendation

- High-priority notification for critically ill patients (visual alert and optional audible notification)

(c) What does the human user do next?

After reviewing the AI recommendation, the triage nurse:

1. Confirms the patient's clinical assessment.
2. Reviews the recommended ESI level and supporting information.
3. Accepts or overrides the recommendation using professional clinical judgement.
4. Records a reason if the recommendation is overridden.
5. Assigns the final ESI level.
6. Prioritises the patient in the Emergency Department queue.
7. Initiates immediate escalation for high-acuity patients when required.

Integration Notes

- Integrates with the **Mercer General Hospital Electronic Health Record (EHR)** system.
- Receives patient registration information automatically after check-in.
- Connects to digital vital-sign monitoring devices to reduce manual data entry.
- Stores AI recommendations, nurse decisions, timestamps, and override reasons in the hospital audit log.
- Uses secure user authentication, encrypted communication, and role-based access control to protect patient information.
- Designed to fit into the existing Emergency Department triage workflow without increasing documentation time.